SQL Mastery Drill: Batch 1

Foundations, DQL, and DML (Q1 - Q20)

BASICS & RETRIEVAL (DQL)

1. What is SQL and why is it used?

"SQL stands for Structured Query Language. It is the standard language used to communicate with Relational Database Management Systems (RDBMS). We use it to store, retrieve, and manage data efficiently in tables."

2. Explain the 'SELECT' command.

"The SELECT command is used to fetch data from a database. You can select specific columns or all columns using the asterisk (*)."

```
SELECT name, category FROM Leads;
```

3. How do you filter records using the 'WHERE' clause?

"The WHERE clause is used to extract only those records that fulfill a specific condition. For example, filtering leads by a specific niche."

```
SELECT * FROM Leads WHERE category = 'Doctor';
```

4. What is the difference between 'AND' and 'OR' operators?

*"AND displays a record if **all** conditions are true. OR displays a record if **any** of the conditions are true."*

5. How do you sort results in SQL?

"We use the ORDER BY keyword. By default, it sorts in ascending order (ASC), but we can specify DESC for descending."

```
SELECT * FROM Leads ORDER BY created_at DESC;
```

DATA MODIFICATION (DML)

6. How do you add new data to a table?

"We use the INSERT INTO statement. You specify the table name and the values to be inserted."

```
INSERT INTO Leads (name, phone) VALUES ('Oviya', '9176614932');
```

7. Explain the 'UPDATE' statement.

*"UPDATE is used to modify existing records. **CRITICAL:** Always use a WHERE clause with UPDATE, or all records in the table will be changed!"*

```
UPDATE Leads SET status = 'Contacted' WHERE id = 5;
```

8. How do you remove records from a table?

"We use the DELETE statement. Similar to UPDATE, a WHERE clause is essential to avoid deleting all data."

```
DELETE FROM Leads WHERE id = 10;
```

CONSTRAINTS & KEYS

9. What is a Primary Key?

*"A Primary Key is a column (or set of columns) that **uniquely identifies** each row in a table. It cannot contain NULL values and must be unique."*

10. What is a Unique Constraint?

"It ensures that all values in a column are different. Unlike a Primary Key, a column with a Unique constraint can accept a NULL value (though only one in some databases)."

11. What is 'NOT NULL'?

"It's a constraint that prevents a column from having a NULL value. This is used to ensure that critical data, like a Lead's name, is always provided."

12. Explain the 'DEFAULT' constraint.

"It provides a default value for a column if no value is specified during an INSERT. For example, setting the 'status' of a new lead to 'Pending' automatically."

WILDCARDS & PATTERNS

13. How do you search for patterns using the 'LIKE' operator?

"We use wildcards like % (represents zero or more characters) and _ (represents a single character). To find names starting with 'O':"

```
SELECT * FROM Leads WHERE name LIKE 'O%';
```

14. What is the 'IN' operator?

"The IN operator allows you to specify multiple values in a WHERE clause. It's a shorthand for multiple OR conditions."

```
SELECT * FROM Leads WHERE category IN ('Doctor', 'Lawyer');
```

15. How do you find values within a range?

"We use the BETWEEN operator. It selects values (including numbers, text, or dates) within a given range."

```
SELECT * FROM Leads WHERE id BETWEEN 1 AND 100;
```

ADVANCED RETRIEVAL BASICS

16. What does 'SELECT DISTINCT' do?

"It is used to return only distinct (different) values. It's useful for finding a list of unique categories in your LeadGen table."

17. How do you limit the number of records returned?

"In MySQL/PostgreSQL, we use LIMIT. In SQL Server, we use SELECT TOP. This is great for showing just the first 10 results on a dashboard."

18. What are SQL Aliases?

"Aliases are used to give a table or a column a temporary name. We use the AS keyword. This makes column names more readable in the final result."

19. How do you check for missing values (NULL)?

"We use 'IS NULL' or 'IS NOT NULL'. You cannot use '=' to check for NULL."

```
SELECT * FROM Leads WHERE phone IS NULL;
```

20. What is the difference between DELETE and TRUNCATE?

*"DELETE is a DML command that can be rolled back and can use a WHERE clause. TRUNCATE is a DDL command that deletes **all** rows from a table, is faster, but cannot be rolled back in many databases."*